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Hyper-Dependence on NHEJ Enables Synergy Between DNA-PK Inhibitors and Low-Dose Doxorubicin in Leiomyosarcoma

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Abstract

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Authors contributions

J.A.F. and A.M-E. conceived and designed the studies. A.M-E., J.P.N., I.K., M.K., M.Z.L., W-B.O., D.L.T., D.F.P-J., and B.A.L. performed experiments. D.C.G., A.V.T., V.Y.M. and P.J.P. analyzed and interpreted sequencing data. A.M-E., J.P.N., F.T.Z., K.V.B., P.C.G., G.S.C., J.A. and J.A.F. collected and analyzed data. A.M-E., S.G. and J.A.F. contributed models and clinical data. L.H.B., D.E.R., K.V.B., P.J.P. and S.G. provided essential reagents, expertise, infrastructure, and/or conceptual advice. A.M-E., J.P.N., and J.A.F. wrote the manuscript, which was reviewed, edited, and approved by all co-authors.

Conflict of interest disclosure statement

Jan Philipp Novotny owns stock in Onxeo. Daniel F. Pilco-Janeta has received travel grants from Pfizer, Roemmers, Roche, and Bristol-Myers Squibb. Frank T. Zenke and Joachim Albers are employees of the healthcare business of Merck KGaA, Darmstadt, Germany. Glenn S. Cowley is currently an employee of Therapeutics Discovery, Janssen R&D, LLC. S. George reports grants and personal fees from Blueprint Medicines (research support), Deciphera Pharmaceuticals (research support), and ARIAD (research support), as well as personal fees from Bayer, Wolter Kluwers Health (royalties), MORE Health, Research to Practice, Eli Lilly, AstraZeneca, Novartis, and other from Abbott Labs outside the submitted work. The other authors declare no potential conflicts of interest.

Purpose: Leiomyosarcoma (LMS) is an aggressive sarcoma for which standard chemotherapies achieve response rates under 30%. There are no effective targeted therapies against LMS. Most LMS are characterized by chromosomal instability (CIN), resulting in part from *TP53* and *RB1* co-inactivation and DNA damage repair defects. We sought to identify therapeutic targets that could exacerbate intrinsic CIN and DNA damage in LMS, inducing lethal genotoxicity.

Experimental design: We performed clinical targeted sequencing in 287 LMS and genome-wide loss-of-function screens in 3 patient-derived LMS cell lines, to identify LMS-specific dependencies. We validated candidate targets by biochemical and cell-response assays in vitro and in 7 mouse models.

Results: Clinical targeted sequencing revealed a high burden of somatic copy number alterations (median fraction of the genome altered=0.62) and demonstrated homologous recombination deficiency signatures in 35% of LMS. Genome-wide shRNA screens demonstrated *PRKDC* (DNA-PKcs) and *RPA2* essentiality, consistent with compensatory non-homologous end joining hyper-dependence. DNA-PK inhibitor combinations with unconventionally low-dose doxorubicin had synergistic activity in LMS *in vitro* models. Combination therapy with pepsertib and low-dose doxorubicin (standard or liposomal formulations) inhibited growth of 5 of 7 LMS mouse models without toxicity.

Conclusion: Combinations of DNA-PK inhibitors with unconventionally low, sensitizing, doxorubicin dosing showed synergistic effects in LMS *in vitro* and *in vivo* models, without discernable toxicity. These findings underscore the relevance of DNA damage repair alterations in LMS pathogenesis and identify dependence on NHEJ as a clinically actionable vulnerability in LMS.

Keywords

Leiomyosarcoma; DNA repair; NHEJ; Cancer dependencies; genomic instability

Introduction

Leiomyosarcomas (LMS) are malignant mesenchymal neoplasms defined by smooth muscle differentiation. LMS may arise anywhere in the body with predilection for the uterus, retroperitoneum, and large blood vessels (1). Complete surgical resection remains the sole curative approach in the absence of effective medical therapies. Thus, locally advanced and metastatic LMS are treated with palliative intent. Genotoxic agents, such as anthracyclines and other topoisomerase II inhibitors, are the backbone of LMS medical treatment and achieve overall response rates of only 25–30%, associated with substantial (dose-limiting) toxicity (2,3). Progression free survival (PFS) remains dismal regardless of therapy, with a median PFS of 4–6 months in most recent LMS trials (1). LMS are molecularly heterogeneous and are characterized by marked chromosomal instability (CIN), featuring aneuploid genomes with numerous copy number alterations and complex rearrangements (4,5). *TP53* and *RB1* genomic inactivation co-occur in most LMS as early events in tumor development, fostering CIN (4,6). *PTEN* and *ATRX* genomic alterations are also frequent in LMS, and further contribute to CIN during LMS progression (4,6).

CIN promotes adaptative oncogenic traits such as intratumoral heterogeneity, resistance to cytotoxic chemotherapy, and increased metastatic potential (7). Unfettered CIN, however, leads to accumulation of genotoxic DNA damage and cell death (8). CIN often manifests as erroneously repaired DNA double strand breaks, which can arise from endogenous or exogenous sources, such as DNA damage repair dysregulation or genotoxic chemotherapy (9). Effective DNA damage repair mechanisms are essential in many cancers for constraining what would otherwise be intolerable levels of double strand breaks and other lethal forms of DNA damage. Double strand break repair relies particularly on two complementary pathways, homologous recombination (HR) and non-homologous end joining (NHEJ), which constrain DNA damage and CIN in a cell context-dependent manner (10,11). Preclinical data suggest that DNA damage and CIN in LMS result, in part, from HR deficiency (4,5,12), providing a potential therapeutic vulnerability that could render LMS cells sensitive to PARP inhibition. This hypothesis is currently being explored in early phase LMS clinical trials (4,5,13–15). However, the prevalence of HR deficiency in LMS remains unclear, as does the relevance of intrinsic genomic instability and other DNA repair pathways that could be therapeutically exploited (15).

We hypothesized that features of compromised genomic integrity in LMS, including *TP53* and *RB1* co-inactivation and CIN, create biologic dependencies that might be therapeutically leveraged, beyond HR deficiency. To test this hypothesis, and to identify therapeutic opportunities to maximize DNA damage in LMS, we analyzed clinical-grade genomic sequencing data from 287 LMS and performed genome-wide loss of function screens on representative patient-derived LMS models. These studies underscore the relevance of DNA damage repair alterations in LMS pathogenesis and demonstrate that LMS critically depends on NHEJ to offset intrinsic and exogenous sources of genomic instability. These findings further suggest that the association of CIN and NHEJ dependence creates a therapeutic opportunity to maximize the deleterious effects of NHEJ inhibition by co-administration of unconventionally low doses of anthracyclines, thereby minimizing the risk of systemic toxicity to LMS patients.

Methods

Genomic characterization (Oncopanel series, n=287)

Targeted sequencing was performed using the Brigham and Women's Hospital OncoPanel clinical sequencing platform (16) on samples from 287 LMS patients, of which 27, 63 and 166 samples were sequenced with OncoPanel versions 1, 2 and 3 with a total coverage of 1.67, 1.94 and 3.49 Mb, respectively. This is an institutional series of cases, selected after diagnostic confirmation by expert sarcoma pathologists, enriched for clinically advanced cases in a tertiary referral cancer center. The cell line models and the PDX model were sequenced with OncoPanel v3. The gene lists are provided in Supplementary Table S1. Comparison sample sets included 62 gastrointestinal stromal tumors (mesenchymal tumor with myogenic biology and stable genome); and 103 breast and ovarian carcinomas with frequent HRD genomic features. This study was performed in accordance with the U.S. Common Rule after IRB approval (BWH IRB protocols 2016P001976 and

2019P001366, DFCI IRB protocol 11–104). Written informed consent was obtained from participants prior to sequencing.

Single nucleotide variant (SNV) and indel calls (n=251)

SNVs and indels were identified by Mutect2 (17) [<https://www.biorxiv.org/content/10.1101/861054v1.full>]. A panel of normal samples sequenced with the same OncoPanel version were used to identify and filter batch-specific sequencing artefacts. Potential germline variants were tagged based on whether they were found in the normal panel or in common variant databases (gnomAD (RRID:SCR_014964) (18) and ExAc (RRID:SCR_004068) (19)) and whether they were filtered out as potential germline variants by Mutect2. The putative somatic variants remaining after this filtering were required to have at least 20 reads supporting the variant, 5 variants supporting an alternate allele, and minimum of 10% allelic fraction. Mutations that are pathogenic based on ClinVar (RRID:SCR_006169) (20), those tagged as damaging by both SIFT (RRID:SCR_012813) (21) and PolyPhen (RRID:SCR_013189) (22), and truncating somatic mutations are shown in Supplementary Table S3.

Sig3 assignments with SigMA

Sig3 status was determined using the SigMA algorithm (23) and an inclusive set of both synonymous (not included in Supplementary Table S3) and nonsynonymous mutations identified by Mutect after germline filtering. A gradient boosting classifier for LMS OncoPanels was developed using training panels simulated from 81 whole genome-sequenced (WGS) LMS samples from ICGC (International Cancer Genome Consortium). This was done by down-sampling ICGC data to the sequencing coverage of the OncoPanels and readjusting the number of SNVs to account for read depth differences between the WGS and OncoPanel data. Signatures were assigned to WGS data with a standard analysis approach that consists of non-negative matrix factorization, matching to the catalog, and refitting. WGS was clustered with hierarchical clustering using signature exposures. Clusters centroids served as expected probability distributions and the clusters that contained samples where signature 3 was assigned were used to calculate Sig3 likelihood. In simulated panels, the likelihood, cosine similarity and signature exposures with non-negative least squares were calculated. These features in simulated panels were used to train a gradient boosting classifier with 5-fold cross validation, and the Sig3+ samples identified based on the analysis of WGS data were used as truth tags. Here, Sig3+ denotes samples with a SigMA score larger than the threshold point that provides 73% sensitivity at 10% FPR in a test cohort of simulated data. The classifier trained on the simulated dataset was then used to predict Sig3 scores in OncoPanel data and determine Sig3+ samples (See Supplementary Tables S3 and S4 for the results in LMS tumor samples and models, respectively).

Copy number alterations and fraction of genomes altered

LMS purity and ploidy were calculated with PureCN (purity is taken to be 1 for cell lines). Using the PureCN results for purity and potential germline variants by Mutect2, copy number alterations were estimated with CNVkit (24) and FACETS algorithms (25) using unmatched normals. FACETS algorithm has been previously used with matched normals (25), and we utilized this algorithm to compare FGA values in our dataset with the external

validation dataset from AACR's project GENIE. Based on our simulations, in the absence of matched normals, FACETS did not identify small CNAs as effectively as CNVkit. FGA was defined as the fraction of the genome with absolute log2 ratios greater than 0.2, as previously published (26). FGAs were higher in Sig3+ samples with CNVkit and not with FACETS ($p = 0.088$, t-test). FGA values with CNVkit can be found in Supplementary Tables S3 and S4 for LMS tumor samples and models, respectively. We identified small deletions and duplications based on whether they had larger or smaller copy number with respect to the arm level copy number calculated as the weighted median of all the segments belonging to that chromosome arm in order to factor out chromosome arm level aneuploidy. The FGA values for CNVkit correlated with small duplications ($p < 0.0001$, $r = 0.46$, Pearson correlation) and not FACETS ($p = 0.43$, $r = 0.07$). Small deletions and duplications were evaluated in v3 and v3.1 OncoPanels because these had sufficient SNP loci (compared to the v1 and v2 OncoPanels) to enable the analyses.

Cell lines and PDX

LMS03, LMS04, LMS05, and GIST882 cells were established before 2007 from primary clinical specimens in the laboratory of Jonathan A. Fletcher at Brigham and Women's Hospital. LMS03 (RRID:CVCL_5H98) was derived from a diaphragmatic metastasis of soft-tissue LMS; LMS04 (RRID:CVCL_5H99) from a retroperitoneal metastasis of uterine LMS with homozygous deletions including *TP53*, *RB1*, and *PTEN*; LMS05 (RRID:CVCL_5I00) from a primary thigh LMS; and GIST882 (RRID:CVCL_7044) from an untreated primary GIST with homozygous exon 13 KIT mutation (p.K642E) (27–29). LMS03, LMS04 and LMS05 cells were maintained in RPMI 1640 (Invitrogen) supplemented with 10% fetal bovine serum (FBS; Invitrogen). GIST882 cells were maintained in IMDM (Invitrogen) supplemented with 15% FBS. All cell lines were supplemented with 100 U/ml penicillin/streptomycin and 4 mM L-glutamine (Invitrogen) and were maintained at 37 °C in 5% CO₂. LMS^{PDX1} was established in the laboratory of Ewa Sicinska at Dana-Farber Cancer Institute, from fresh tissue from a surgically resected uterine LMS in a 52-year-old female, which was implanted subcutaneously into ~6-week-old female nude mice (NU/NU; Charles River 143 Laboratories), and has been described before as LMS4 (30). Cell lines were periodically authenticated (every ~20 passages) by genomic analyses.

Primary shRNA screens

Development and applications of the 98K lentiviral shRNA pooled library from the Broad Institute's Genetic Perturbation Platform have been described previously (31,32). In brief, LMS cells were infected with a pool of 107,523 shRNA lentiviruses targeting 17,098 genes and subjected to puromycin selection. Replicates of 30 million infected LMS03, LMS04 and LMS05 cells were established after the infections and allowed to proliferate independently for ~16 doublings. Genomic DNA was isolated from final harvests of cultured cells for shRNA amplification and massively parallel sequencing as described previously. (31) The 101,626 shRNAs were ranked by their relative depletion from the cell pool, and the corresponding 16,701 genes were then scored according to the rank of the second-most depleted shRNA (out of ~5 shRNAs targeting each gene), using the GENE-E

program (<http://www.broadinstitute.org/cancer/software/GENE-E/download.html>). Essential genes are depleted and rank on the top of the distribution.

In vitro / in vivo knockdown validations

The following lentiviral constructs encoding shRNA-specific sequences targeting *PRKDC* (TRCN0000195491 and TRCN0000006255) and *RPA2* (TRCN0000231924 and TRCN0000231921) transcripts on pLKO.1puro or pLKO_005 backbones were from the GPP library. Lentivirus preparations and infections were performed as described previously. Cells were lysed for immunoblot analysis at 6-, 10-, and 20-days post-infection. Whole-cell lysates, electrophoresis and immunoblotting were carried out as described previously (33) with the following antibodies and dilutions: anti-DNA-PKcs (#4602S, Cell Signaling), 1:1000; anti phospho-DNA-PKcs Ser2056 (#68716S, Cell Signaling), 1:1000; anti-RPA32/RPA2 (#52448S, Cell Signaling), 1:1000; anti-phospho-RPA32/RPA2 Ser8 (E5A2F, #54762S, Cell Signaling), 1:1000, anti-phospho-H2AX Ser139 (#9718S, Cell Signaling), 1:200; PARP (9532, Cell Signaling), 1:1000; Cleaved PARP (#9541S, Cell Signaling), 1:1000; anti-CHK1 (#8408, Cell Signaling), 1:1000; anti-phospho-CHK1 Ser345 #2348, Cell Signaling), 1:1000; and anti-GAPDH (G8795, Sigma), 1:5000.

Deep learning-based cell culture confluence analysis

Representative images of LMS04 cell cultures were captured three times per week on a Leica phase contrast inverted microscope using a SPOT RT2520 camera (SPOT imaging, Sterling Heights, Michigan, USA). Segmentation was performed with the generalist deep learning-based segmentation algorithm Cellpose v0.6 (34), using multiple segmentation runs with increasing cell size diameter parameter (30–120px) to capture all cells. Cell masks were exported, and cell density was measured using Adobe Photoshop v23.1.0, as a fraction of pixels covered by the segmentation mask of total image area of 1600×1200 pixels.

Drug sensitivity in vitro assays

Viability studies were carried out using the CellTiter-Glo luminescent assay (Promega, Madison, WI). Cell proliferation was measured by BrdU incorporation, using a cell proliferation ELISA kit (Roche). Cells were plated at 5 to 10,000 cells per well in 96-well flat-bottomed plates (Falcon, Lincoln, NJ), cultured in serum-containing media for 24h, and then incubated for 6–14 days with increasing doses of doxorubicin HCl (D-4000, LC Laboratories), peposertib (CT-M3814, Chemietec), AZD7648 (LC Laboratories) or DMSO-only solvent control. Luminescence was measured with a Veritas Microplate Luminometer (Turner Biosystems, Sunnyvale, CA), and the data were normalized to the DMSO-only control group. All experimental points were measured in triplicate wells for each plate and were replicated in at least two plates. Logistic regression curves, IC50 calculations, and result plots were generated with GraphPad Prism (9.3.0). Drug combination efficacy was evaluated by the Bliss independence model, according to which positive scores indicate synergy. Bliss scores were calculated across a matrix of drug doses using SynergyFinder (RRID:SCR_019318) (35).

LMS in vivo studies

In vivo studies incorporating standard (non-liposomal) doxorubicin were performed according to the approved protocols of the Dana-Farber Cancer Institute's Institutional Animal Care and Use Committee (IACUC) using 8-week-old female NRG (NOD rag gamma) mice (The Jackson Laboratory, ME). For tolerability studies, body weights of five mice per treatment or vehicle arm were obtained daily. For efficacy studies, tumor fragments of LMS04 xenograft and LMS-PDX1 models were dipped in Matrigel then implanted subcutaneously in the right flank of the NRG mice. When tumors reached 150–250 mm³, mice were randomized to vehicle, doxorubicin (purchased from the DFCI pharmacy), peposertib (synthesized by the healthcare business of Merck KGaA, Darmstadt, Germany), or combination doxorubicin and peposertib treatment (n = 8–9 mice per arm). Doxorubicin (0.5–1 mg/kg) was administered intravenously once a week, whereas peposertib (100 mg/kg) or vehicle (0.5% Methocel with 0.25% Tween 20 in 300 mM sodium citrate, pH 2.5) were administered by gavage twice a day. Tumor volume (volume = length*width²/2; measured with calipers) and body weight were measured twice a week. The primary endpoint was total tumor volume larger than 2,000 mm³.

In vivo studies with liposomal doxorubicin (Doxil) were coordinated by the healthcare business of Merck KGaA, Darmstadt, Germany, and performed at Champions Oncology according to the guidelines of the Institutional Animal Care and Use Committee (IACUC) of Champions Oncology. For these PDX studies, stock mice were bilaterally implanted with fragments from each of the 5 Champions TumorGraft[®] models CTG-1006, CTG-1517, CTG-1082, CTG-1005 and CTG-1079. After the tumors reached 1000–1500 mm³, they were harvested, and the tumor fragments were implanted s.c. in the left flank of the female study mice. Tumor growth was monitored twice a week using digital calipers, and the tumor volume (TV) was calculated using the formula (0.52 × [length × width²]). When the TV reached approximately 150–300 mm³, animals were matched by tumor size and assigned into vehicle control or treatment groups (n = 6–8/group), and dosing was initiated on d0 up to d60 or until mean TV in one group reached 1500 mm³. Peposertib was formulated in vehicle (0.5% Methocel, 0.25% Tween20, 300 mmol/L sodium citrate buffer, pH 2.5) and administered orally at described doses. Pegylated liposomal doxorubicin was injected into the tail vein once a week at the indicated dose. Tumor size and body weight were measured twice a week. Histopathological and genomic analyses for these five models were performed at Champions Oncology, and data was reviewed at BWH and DFCI.

Data Availability

The shRNA screen data are available upon request from the corresponding author. Anonymized clinical targeted sequencing data and associated metadata that support the findings of this study are publicly available in GENIE, through cbiportal; sequencing raw files are not available due to privacy restrictions on clinical data. Other data generated in this study are available within the article and its Supplementary Data files.

Results

LMS targeted sequencing reveals widespread DNA damage and HRD features

LMS genomic profiles were obtained by targeted sequencing of 287 cases using an institutional clinical sequencing platform (Oncopanel) that interrogates 447 cancer-related genes (Supplementary Table S1). This LMS group was characterized by genomic complexity, aneuploidy, and a high burden of somatic copy number alterations (SCNAs). The fraction of genome altered, a measurement of the burden of copy number and structural alterations (36), was high in LMS, at a median of 0.62. This is similar to an internal Oncopanel dataset of breast and ovarian carcinomas enriched for HRD cases (0.67, n=103) and significantly higher than a comparator set of gastrointestinal stromal tumors (GIST) (0.41, n=62; $p < 0.0001$, t-test) (Fig. 1A). The LMS genomic findings were corroborated by independent analyses on a publicly available dataset that includes 305 LMS (GENIE, v9.0 (37)) which confirmed that LMS has significantly higher fraction of genome altered than 44,724 non-LMS cancers in GENIE ($p < 0.0001$, t-test) (Supplementary Fig. S1A, and Supplementary Table S2). Undifferentiated pleomorphic sarcoma and osteosarcoma are two other major subtypes of sarcomas which, like LMS, feature *TP53* and *RB1* co-mutation, high histologic grade, and cellular pleomorphism; these sarcoma subtypes in GENIE were also characterized by high fraction of the genome altered, comparable to LMS (Supplementary Fig. S1B). Arm-level aneuploidy, a different metric of SCNA burden, was also significantly higher in LMS (median 18) and breast and ovarian carcinomas (median 24) than in GIST (median 13) (Fig. 1B). Of note, the Oncopanel profiles demonstrated that copy number alterations in LMS were significantly enriched for small (<30kb) deletions, compared to other cancer types (Fig. 1C); mutational signatures with abundant small deletions have been mechanistically associated with homologous recombination deficiency and active NHEJ (38).

Mutational signature analysis on the OncoPanel sequencing dataset using SigMA (Signature Multivariate Analysis (23)) identified the presence of the HRD-associated mutational signature 3 (Sig3) in 35% (58/166) of the LMS (Figure 1D). A significantly higher fraction of the genome was altered in Sig3+ LMS samples ($p < 0.001$, t-test; Fig. 1E), and in LMS with inactivating *ATR* mutations ($p < 0.001$, t-test; Fig. 1F). Enrichment for small (<30Kb) deletions in LMS was independent of Sig3 status and was associated with deleterious or likely deleterious mutations in *POLH*, *POLD1*, *FANCM* and *XRCC1* (Supplementary Fig. S2), all of which are involved in DNA double strand break repair at different phases of the cell cycle. Taken together, these results demonstrate that LMS genomes have an unusually high burden of SCNAs, mainly <30Kb deletions, which may reflect dependence on NHEJ as a compensatory response to the observed HRD. Mutational analyses are summarized in Supplementary Fig. S3 and Supplementary Table S3.

LMS models developed in our laboratory and used for this report captured the mutations, mutational signatures, and fraction of genome altered profiles observed in the 287 clinical LMS samples (Supplementary Table S4). Fraction of genome altered scores for LMS03, LMS04 and LMS05 cells, and the LMS^{PDX1} PDX were 0.725, 0.755, 0.195, and 0.28, respectively. LMS04, LMS05 and LMS^{PDX1} are HR deficient (Sig3+), while LMS03 is

HR proficient (Sig3-). The GIST882 GIST cell line (HR proficient) was used as a non-LMS comparator because this GIST model, (39) like most LMS, features *TP53* and *RB1* inactivation but (unlike most LMS) is genomically stable (40). Further, transdifferentiation has been demonstrated between interstitial cells of Cajal and smooth muscle, which are the cell lineage counterparts for GIST and LMS, respectively (41,42).

Genome-wide loss-of-function screens identify *PRKDC* as a critical dependence in LMS cells

To identify genes essential for LMS cell proliferation and survival, we performed a negative selection genome-wide shRNA screen in the LMS03, LMS04, and LMS05 patient-derived LMS cell lines (Supplementary Table S4). These screens used a lentiviral library of 98,000 shRNAs targeting 16,700 genes, developed at the Broad Institute (Fig. 2A, details in the methods section). To identify dependencies specific to LMS cells, gene ranks were compared with similar screens performed in GIST882 and with a reference dataset of commonly essential genes in 712 cell lines analyzed within the Dependency Map Project (43).

Analysis of the top 5% (n=835) essential genes in the three LMS cell lines identified 100 genes specifically essential to LMS, whereas the remaining 735 were commonly essential genes. Pathway analysis of LMS essential genes identified significant enrichment of the biological processes “cell cycle”, “DNA repair” and “double strand break repair” (FDR q-values $2e^{-3}$, $4.31e^{-3}$, and $3.86e^{-3}$, respectively). The NHEJ-associated gene *PRKDC*, which encodes the catalytic subunit of DNA-PK, was the highest-ranking DNA repair-related gene in each of the LMS lines. Further, *RPA2*, the direct substrate of DNA-PK in NHEJ pathways, ranked in the top 5% of essential genes in each of these lines (Fig. 2B). shRNA-mediated *PRKDC* knockdown resulted in up to 80% reduction in LMS cell growth after 7 days (Fig. 2C and 2D and Supplementary Fig. S4).

Pharmacologic inhibition of DNA-PK sensitizes LMS to unconventionally low dose of doxorubicin

We then interrogated if pharmacologic inhibition of DNA-PK with peposertib (44) replicated the effects of shRNA-mediated *PRKDC* downregulation in LMS cells. Treatment with peposertib resulted in dose-dependent reduction of LMS cell viability at day 6 (IC₅₀ 0.7–1.29μM). In contrast, GIST882 required concentrations ~100x higher to minimally reduce cell viability (Fig. 3A).

To maximize the effects of DNA-PK inhibition on LMS cells, we designed a combination strategy adding doxorubicin treatment to induce additional double strand breaks and increase cumulative DNA damage. We sought to model an approach that would minimize toxicity in the clinic by combining low dose peposertib (400nM) with unusually low doxorubicin concentrations that, by design, would lead to incremental, cumulative, genotoxic effects on LMS cells rather than immediate toxicity. To this end, we initially performed short-term doxorubicin dose-response experiments in LMS and GIST cells: doxorubicin 2nM resulted in cell viability values of 70–100% at day 6 for LMS cells (Fig. 3B). Likewise, 2nM doxorubicin did not have discernable biochemical effects in LMS cells at 24–48h,

with no induction of pRPA2, pH2AX or cleaved caspase detectable by immunoblotting (Supplementary Fig. S5). Single agent peposertib (400nM) for 6 days reduced LMS04 and LMS05 cell viability to 67% and 77%, respectively, of untreated controls.

The combination of peposertib with 2nM doxorubicin reduced cell viability to 38% (Fig. 3C); similar results were observed in HR-proficient LMS03 cells with 400nM peposertib and 10nM doxorubicin (Fig. 3C), whereas this combination treatment had minimal impact on the GIST882 comparator cells. Biochemically, treatment of LMS04 cells with peposertib and doxorubicin combination for 48h resulted in induction of cleaved PARP, indicating apoptotic pathway activation (Fig. 3D).

The LMS cells surviving the combination of peposertib and doxorubicin were large, non-refractile, and non-dividing, consistent with senescence (Supplementary Fig. S6 and Supplementary Fig. S7, Days 6 and 13) (45). Due to this increased cell size, the CTG-based assays used herein, which measure ATP content as a surrogate for cell viability, underestimate the combination drug impact on cell growth (46). Therefore, we also performed drug response studies using a proliferation assay (BrdU incorporation). These studies demonstrated that the combination of DNA-PKi and low-dose doxorubicin synergistically inhibits LMS cell proliferation, as determined by positive Bliss scores in LMS03 (max: 39.14; average: 13.64), LMS04 (max: 38.77; average: 16.88), and LMS05 (max: 48.41; average: 21.69) (Supplementary Fig. S8). These inhibitory effects were confirmed using an alternative DNA-PKcs inhibitor, AZD7648 (Supplementary Fig. S9).

As hypothesized, prolonged treatment with low dose peposertib combined with unconventionally low dose doxorubicin produced cumulative effects on LMS cells: the viability of LMS cells at day 14 was significantly reduced in comparison with day 6 and with monotherapy, and in comparison with the same treatments in GIST cells (Fig. 4A). Sensitivity to combinations of peposertib 400nM with low-dose doxorubicin was higher for long-term than short-term treatment and correlated with HRD status. The HRD-deficient LMS cell lines, LMS04 and LMS05, were highly sensitive (3% and 18% cell viability, respectively, with 2nM doxorubicin) whereas LMS03 was less sensitive (75% and 5% viability at 2nM and 10nM doxorubicin, respectively). Longer-term LMS04 cultures were not viable under combination treatment, with death of all cells after 20 days, in contrast to peposertib or doxorubicin monotherapy (Fig. 4B–C, Supplementary Fig. S6). These findings support the hypothesis that DNA-PK inhibition can leverage even unconventionally low dosing of a DNA-damaging agent, leading to cell death by cumulative genotoxicity. Notably, discontinuation of drug treatment led to full recovery and re-growth of LMS04 cells after peposertib or doxorubicin monotherapy treatment but not after treatment with the peposertib and doxorubicin combination.

***In vivo* efficacy of peposertib and low dose doxorubicin in LMS**

To evaluate efficacy and toxicity of combinations of peposertib with unconventionally low-dose doxorubicin *in vivo*, we treated LMS04 mice xenografts with 20% of the commonly applied doxorubicin dose in mice (1mg/kg BW qw) in combination with peposertib at a standard dose of 100mg/kg BID (Fig. 5). This combination was well tolerated, as evidenced by unaltered weight and behavior. Peposertib treatment led to 71% reduction of xenograft

growth, compared to xenografts in control mice treated with vehicle, while co-treatment with doxorubicin resulted in 96% reduction in growth (Fig. 5A) with highest tumor regression achieved at day 8. To assess efficacy of the pposertib/doxorubicin combination in a model with potential resistance to doxorubicin, we established a mouse PDX from a heavily pre-treated LMS patient who progressed clinically on both doxorubicin and liposomal doxorubicin (LMS^{PDX1}). Treatment of mice carrying LMS^{PDX1} with vehicle, pposertib, low-dose doxorubicin, and the combination pposertib/low-dose doxorubicin, demonstrated >45% reduction in tumor growth in the group of mice treated with pposertib and low-dose doxorubicin, which was statistically significant when compared to the vehicle treatment ($p<0.05$, Fig. 5B).

The clinical extended half-life of liposomal doxorubicin, compared to non-liposomal doxorubicin, might permit greater treatment synergies in combination with daily DNA-PKi, albeit with greater risk of toxicity. Therefore, we tested the combination of liposomal doxorubicin with pposertib in 5 LMS PDX mice models. Tumor-bearing mice were treated with pposertib (100mg/kg BID), liposomal doxorubicin (3mg/kg), or the combination of both. This dose of liposomal doxorubicin corresponds (based on Body surface area, [BSA]) to approximately 20–25% of the commonly used dose in humans (40 mg/m²). Genomic and clinicopathological features of these PDX models, which collectively capture the genetic and biological spectrum of LMS, are summarized in Supplementary Table S5. While all animals in the vehicle or monotherapy groups reached the tumor volume endpoint of 2,000 mm³ by day 26, 4/6 animals in the combination group could be kept on treatment until day 70 without reaching the tumor volume endpoint. Even though HR status is not available for all these models, two PDX (CTG-1006 and CTG-1517) with *ATRX* inactivation as evidence of HRD had substantial and prolonged responses to the pposertib and liposomal doxorubicin combination (Fig 5C and 5D). Further, the best response was in CTG-1006, which had biallelic *BRCA2* deletion. Three additional LMS models (CTG-1182, CTG-1079 and CTG-1005) had reduced tumor volume with combination therapy compared to the constituent monotherapies (Supplementary Fig. S10), and this differential growth reached statistical significance in CTG-1182. Overall, the combination of pposertib with low-dose doxorubicin or its liposomal formulation demonstrated activity in 5 of 7 *in vivo* LMS models. Importantly, the combination regimen was well-tolerated (Supplementary Fig. S11).

Discussion

LMS is a genomically unstable sarcoma. Integrative molecular studies have identified genetic lesions in LMS that likely foster this instability: co-inactivation of the tumor suppressor genes *TP53* and *RB1* is a fundamental event in almost all LMS and can be an initiating event (5,6). Subsequent loss of *PTEN* and *ATRX*, which occur, respectively, in 57–71% and 15–24% of LMS, further affect genome integrity (47,48). Additional alterations involve genes related to smooth muscle biology (*DMD* deletions/inactivation and *MYOCD* amplification) or cell cycle regulation (*CDKN2A*, *CDKN2B* or *CDKN1C*) (4–6). Our single-institution evaluations of 287 LMS genomes underscore that the fraction of the genome altered (FGA) and the degree of arm-level aneuploidy are unusually high in LMS, underscoring the genomic complexity of LMS. Notably, among other sarcoma subtypes, only those that share with LMS frequent high-grade pleomorphic features and *TP53* and

RB1 inactivation – osteosarcoma, UPS, and myxofibrosarcoma – showed similarly high burden of somatic copy number alterations, suggesting that these karyotypically complex sarcomas share defects in genome integrity maintenance mechanisms similar to those prevalent in LMS.

Therapeutically, genomic complexity in cancer cells may be exploited with strategies that increase genotoxicity to untenable levels (8,9). Indeed, the mainstays of LMS drug therapy continue to be conventional cytotoxic therapies that target genomic integrity. Unfortunately, the therapeutic benefit in LMS is limited and even in those responding to DNA damaging agents the duration of therapy is constrained by cumulative toxicity to non-neoplastic cells (49). Our studies provide rationale for additional approaches to genomic destabilization that target LMS biologic vulnerabilities, particularly pairing unconventionally low dosing of a DNA damaging agent with drugs targeting DNA-PK as an essential DNA damage repair factor in LMS. In all, these strategies aim to leverage the intrinsic DNA damage in most LMS and can be potentially extended to combinations of DNA damage repair inhibitors, creating opportunities to broaden the therapeutic window by increasing tolerability through selectively targeting genomically-complex cancer cells (50).

LMS genomic complexity and DNA damage result from mechanisms that include whole genome doubling, chromothripsis and, importantly, HR deficiency (HRD) (4,5,12). Mutational signature analyses in our current studies and those reported previously suggest that HRD contributes to the mutational repertoire of 35%–98% of LMS (4,5,12). Preclinical evidence supports that HRD constitutes a targetable vulnerability that renders LMS cells sensitive to synthetic lethal therapeutic approaches involving PARP inhibitors (4,5,13,14). Such approaches are currently being explored in early phase clinical trials in LMS patients, which evaluate combinations of PARPi and DNA-damaging agents at conventional doses (15). Preliminary results and anecdotal reports suggest that a subset of patients with *BRCA1/BRCA2*-mutant HRD LMS derive therapeutic benefit from combinations including PARP inhibition, at the expense of non-negligible toxicities (15,51). Notably, HRD status or mutational inactivation of HR-related genes do not reliably predict response to PARPi combinations in these trials, suggesting that sensitivity to PARPi in LMS is not determined exclusively by HRD, and that other DNA repair defects contribute to intrinsic genomic instability in LMS (15).

Analysis of copy number alterations in our series identified enrichment for deletions of small size (<30 Kb) as a main contributor to the high FGA in LMS. Small deletions within various sequence contexts are characteristic of certain insertion/deletion mutational signatures, namely ID6 and ID8 (38), which have been mechanistically associated with HRD and active NHEJ. Using the machine-learning-based algorithm SigMA, we also identified a significant contribution of signature 3 to the mutational repertoire of 35% of LMS. Signature 3 is characterized by high frequency of deletions up to 50 bp with overlapping microhomology at breakpoint junctions (23). Further, inactivating mutations in HR-related genes were demonstrated in 10% of LMS in this series. Notably, mutations disrupting *ATRX* and genes involved in non-homologous DNA repair (*POLH*, *POLD1*, *FANCM*, *XRCC1*) were associated with higher FGA, suggesting that DNA repair defects other than HRD contribute to LMS genomic instability. Indeed, germline studies identified pathogenic or

likely pathogenic germline alterations in 22/162 LMS patients (13.6%), of which 81% involved DNA damage-response genes (Prof. David Thomas, personal communication, data from (52)), including *BRCA2* (affected in one LMS) and *ERCC2*, *ERCC3*, *FANCC*, *POT1*, *RECQL4*, *WRN*, and *BLM*. These observations suggest that DNA repair defects in LMS are multifaceted and provide rationale for therapeutic approaches exploiting DNA repair-associated vulnerabilities above and beyond synthetic lethal combinations with PARPi.

Results from our genome-wide functional experiments indicate that LMS cells depend exquisitely on *PRKDC* for survival. *PRKDC* encodes DNA-PK, a serine/threonine protein kinase that acts as a molecular sensor for DNA damage and is required for DNA double-strand break repair through NHEJ. The dependence of LMS on *PRKDC* suggests compensatory reliance on NHEJ to manage the high levels of DNA damage, particularly double strand breaks, which are inherent to highly unstable genomes (53). Based on these discoveries, we hypothesize that DNA-PK hyper-dependence is prevalent in LMS, and may be fostered in the context of HRD, given the complementary nature of NHEJ and HRD as the two major classic pathways for DNA double strand break repair (54). Recently published work has shown that HRD LMS cell lines show increased NHEJ activity when subject to endonuclease-induced double strand breaks using a fluorescent reporter assay (4,5); our results support these observations and highlight the extreme dependence of LMS cells on NHEJ at baseline, independent of exogenous sources of DNA-damage. It remains to be determined whether NHEJ hyper-dependence in LMS is primarily a compensation for HRD, and what other factors enhance or mitigate this dependence. These questions underscore the need for continued advances in predictive assays that demonstrate dependencies *vs.* deficiencies in homologous recombination and NHEJ repair pathways. Our analyses of clinical targeted genome sequencing suggests that enrichment for small deletions or presence of inactivating *ATR*X alterations may be useful genomic surrogate biomarkers for NHEJ dependence in LMS. These analyses can be easily incorporated into analytical pipelines of clinical sequencing assays, as we aim to validate in ongoing clinical trials.

Consistent with the dependence of LMS cells on NHEJ, DNA-PK pharmacological inhibition, particularly in combination with low-dose anthracycline, inhibited growth of *in vitro* and *in vivo* LMS models. Indeed, LMS response to the DNA-PK inhibitor, peposertib (44), correlated with HRD. Peposertib is a highly selective oral DNA-PK inhibitor which has been evaluated in a phase 1 trial (55) showing modest single-agent activity with best overall response of stable disease in 12 of 31 patients with unselected tumors. In this trial, the MTD was not reached and the most frequent adverse events were fatigue, nausea/vomiting, and fever. Peposertib activity is now being evaluated in combinations with genotoxic agents or radiation therapy in several cancer types, including rectal cancer [NCT03770689, NCT04068194], glioblastoma [NCT04555577], and other advanced solid tumors [NCT02516813]. These approaches are based on the ability of peposertib to sensitize cells to the effects of radiation or chemotherapy, which are typically administered at full doses, and whose effects are enhanced upon DNA-PK inhibition (56–58). Because previous trials of DNA-PKi combination therapies were hampered by substantial toxicity (59), we modeled approaches in which DNA-PK inhibition was combined with unconventionally low anthracycline dosing. To this end, we defined a dose range of doxorubicin that had minimal short-term biochemical or viability effects in LMS cells yet acted as a sensitizer to the

effects of DNA-PK inhibition in longer-term studies. The anthracycline doxorubicin is the most effective single agent therapy against LMS, although its clinical efficacy is limited by acute and chronic cardiotoxicity. Cardiotoxicity is a class-specific and dose-dependent adverse effect that portends poor prognosis and is potentially lethal (60). There is no effective treatment for doxorubicin cardiotoxicity, so the usual preventive strategy is to limit the cumulative dose to $<450 \text{ mg/m}^2$ (61,62), which typically allows for 6 cycles of treatment over ~6–7 months. We hypothesized that unconventionally low doses of doxorubicin might sensitize LMS to DNA-PK inhibition, thereby leveraging the differential therapeutic index of LMS to NHEJ-i, inducing sufficient double strand DNA breaks to exceed a genotoxicity threshold that triggers cell death. This concept is particularly suited to LMS, given the high levels of endogenous DNA damage and the underlying DNA repair defects. Our results support this hypothesis, demonstrating that unconventionally low doxorubicin dosing sensitizes LMS to pposertib, over time. We observed efficacy of this combination in multiple LMS models, whereas monotherapies with these same drugs had little activity. These observations suggest that reducing the dose of genotoxic agents in combinations with NHEJ inhibitors is a rational strategy to minimize toxicities, while preserving anti-tumor activity. We are testing this concept in a phase 1 dose escalation and dose expansion trial of low dose liposomal doxorubicin combined with pposertib, with embedded correlative studies to identify predictive biomarkers and explore mechanisms of action (NCT05711615). Because this concept could be relevant in other cancers with complex genomes and CIN, we are including other karyotypically complex sarcomas in the dose escalation stage of the trial. Notably, this paradigm might be applicable to combinations of DNA-PK inhibitors with other cytotoxic agents or other DNA damage repair inhibitors, and further pre-clinical studies are warranted to identify additional promising approaches.

Despite an improved understanding of the genetics of LMS, current treatments for patients with advanced LMS remain centered on the antitumor activity of genotoxic chemotherapeutic agents. Therapeutic regimens including PARPi hold promise given the contribution of HRD to LMS genomes and are being investigated (15). The work presented herein supports a novel therapeutic strategy that involves targeting NHEJ, a complementary DNA-repair mechanism critical to LMS cells, while reducing the likelihood of undue toxicity. At present, it is unclear which LMS patients may derive most benefit from DNA-PK inhibition, from PARP inhibition, or from a potential combination of both. Likewise, although chromosomal instability, HRD, and NHEJ dependency likely contribute to pposertib and doxorubicin sensitivity, the relationships between these mechanisms are currently only partially understood. According to our results, quantification of copy number alteration burden using FGA and enrichment for deletions $<30\text{Kb}$ provides means for better identification of patients with NHEJ-dependent LMS, related or not to HRD, who might derive most benefit from DNA-PK inhibition with pposertib. Importantly, a “gold standard” assay for assessing HRD in the clinical setting remains elusive. Biomarkers that rely on varied genomic assessments of HRD features (loss of heterozygosity, telomeric allelic imbalance, large scale transitions, or some combination of these) fail to capture all responders to PARPi (63). It is hoped that genomic signatures including mutational signature 3 and indel signatures ID6 and ID8 will identify a larger proportion of tumors that may benefit from PARPi therapy (63). Interestingly, these signatures likely reflect

the reparative effects of active NHEJ on DNA double strand breaks, and hence may also enrich for tumors hyper-dependent on NHEJ that could benefit from DNA-PKi treatment. These considerations will contribute to the design of future clinical trials for LMS patients, which will require genomic studies, mutational analyses, and innovative evaluations of DNA damage repair pathways to develop useful biomarkers for optimal patient stratification.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Availability

The shRNA screen data are available upon request from the corresponding author. Anonymized clinical targeted sequencing data and associated metadata that support the findings of this study are publicly available in GENIE, through cbiportal; sequencing raw files are not available due to privacy restrictions on clinical data. Other data generated in this study are available within the article and its Supplementary Data files.

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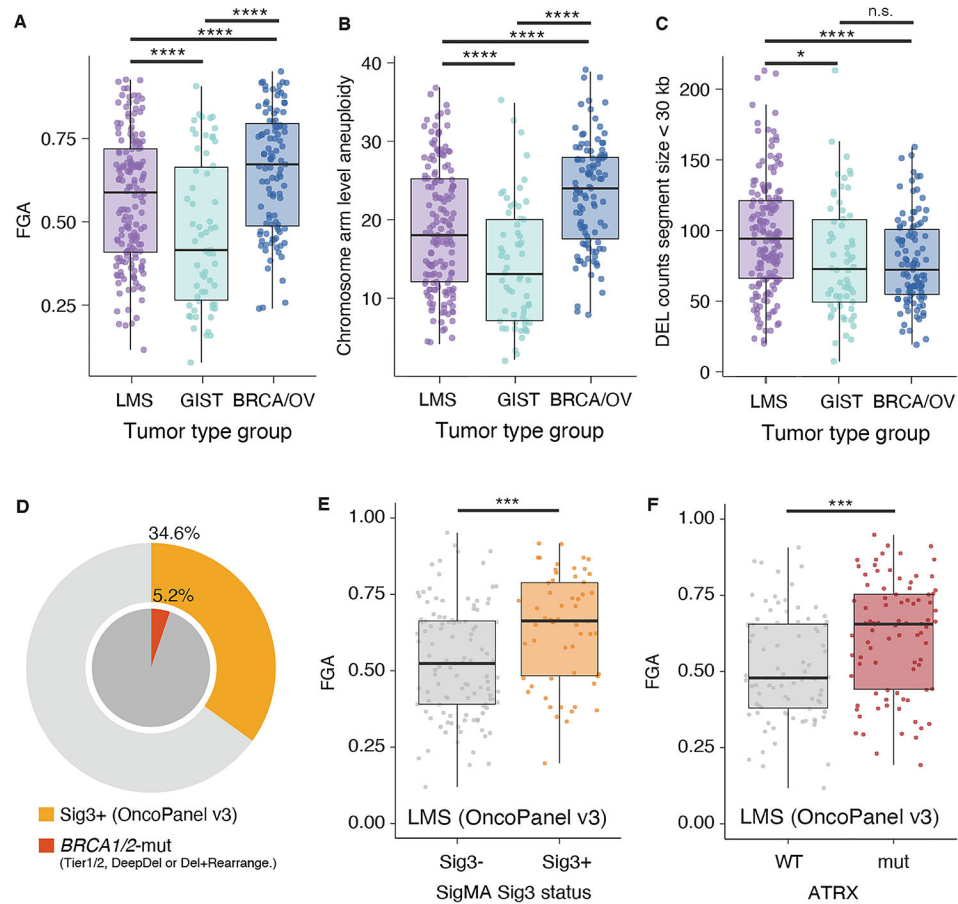
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Statement of Translational Relevance

Leiomyosarcoma (LMS) is one of the most common sarcoma subtypes. The LMS genomic and functional profiles reported herein reveal genomic complexity and hyper-dependence on DNA-PK and non-homologous end joining DNA repair mechanisms. These discoveries define therapeutic opportunities in which ultra-low doses of conventional DNA damaging agents sensitize LMS to DNA-PK inhibitors. This approach leverages intrinsic genomic instability characteristic of LMS and other advanced cancers, and minimizes the substantial toxicities previously seen with combinations of DNA-PK inhibitors and DNA damaging agents.

**Figure 1:**

Prevalent genomic instability in LMS: Fraction of genome altered (FGA) (**A**), chromosome arm-aneuploidy (**B**) and number of deletions <30kb (**C**) in LMS vs. gastrointestinal stromal tumor (GIST, n=62) and breast and ovarian carcinoma (BRCA/OV, n=103). GISTs and LMS are related mesenchymal tumors with myogenic features (39), but the key oncogenic event in most GISTs is constitutive KIT/PDGFR tyrosine kinase activation rather than *TP53* and *RB1* dysregulation. **D**. Fraction of LMS with Sig3+ samples (OncoPanel v3, n=166) and with pathogenic *BRCA1* or *BRCA2* mutations. **E**. FGA values in Sig3+ LMS are significantly higher compared to Sig3- LMS. **F**. FGA values in *ATRX*-mutant LMS are significantly higher than in *ATRX* WT cases. (**** = $p < 0.0001$; *** = $p < 0.001$; n.s.= not significant).

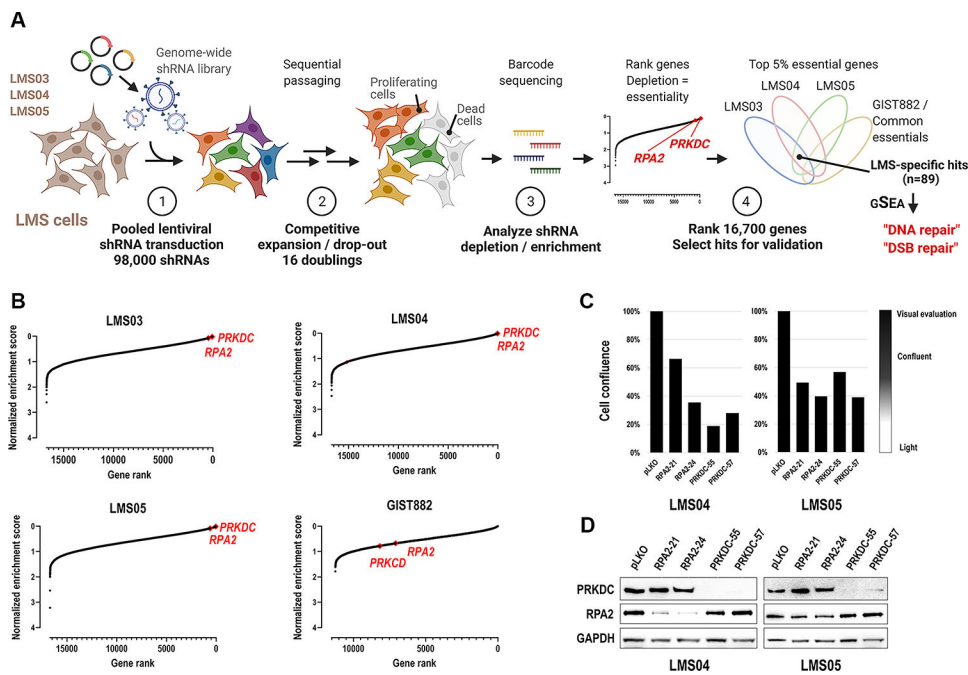


Figure 2: Genome-wide loss of function screens. **A.** Experimental design of shRNA screens in 3 LMS cell lines. **B.** End-point gene ranks (normalized enrichment core) of the screens demonstrate significant depletion of *PRKDC* and *RPA2* in each LMS, reflecting functional dependence. **C.** Computer-assisted cell confluence analysis of LMS04 and LMS05 cultures shows decreased growth after shRNA-mediated knockdown of *PRKDC* and *RPA2* at day 7 (photomicrographs shown in Supplementary Fig. S4) **D.** Knockdown efficiency of 2 independent shRNAs targeting *PRKDC* and *RPA2* in LMS cells. Conditions correspond to the cell cultures quantified in Fig. 2C.

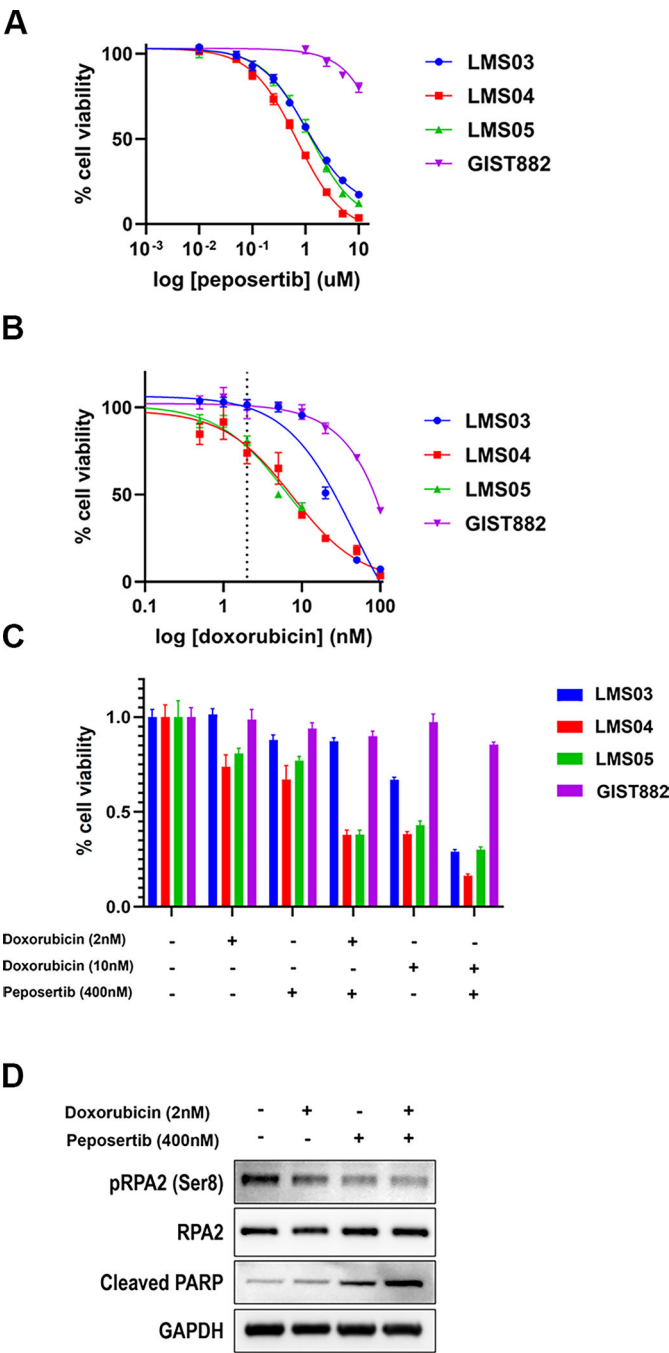


Figure 3: Effects of peposertib, low-dose doxorubicin and combination treatments in LMS. Peposertib (A) and doxorubicin (B) dose-response curves, demonstrating greater inhibition of viability in LMS than in GIST. Note chemoresistance of GIST882 cells, characteristic of this tumor type. C. Combination of peposertib and doxorubicin shows differential activity against HR-deficient LMS04 and LMS05, and HR-proficient LMS03, compared to GIST. D. Biochemical effects of peposertib, doxorubicin, or combination treatments on LMS04 cells

assessed by immunoblotting at 24h: DNA-PKcs inhibition is evidenced by loss of RPA2 phosphorylation and is associated with increased DNA damage (cleaved PARP).

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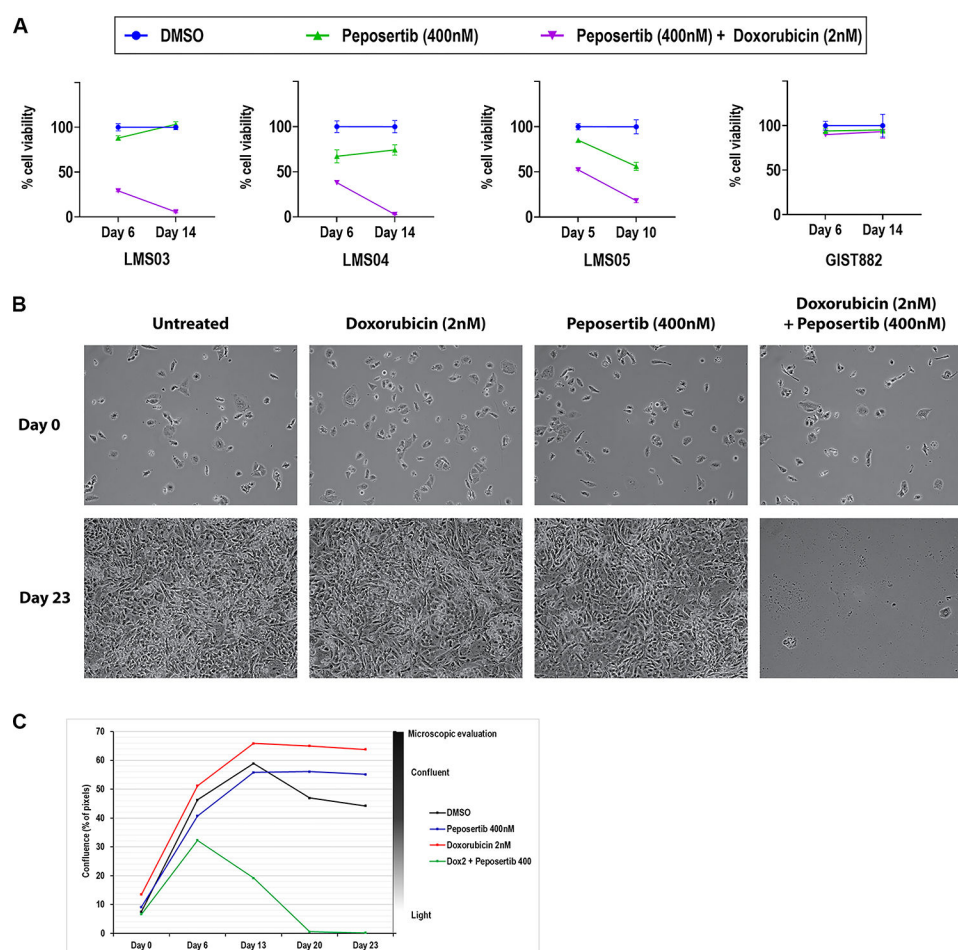


Figure 4: Long-term effect of peposertib (400nM), and combination treatment with low-dose doxorubicin (2nM) in LMS and GIST cells. **A.** Longer exposure to combination treatment results in significantly decreased viability of LMS cells but not GIST cells. **B.** Cell culture morphology of LMS04 cells at day 23, compared to day 0; and **C.** computer-assisted confluence analysis of a time-course experiment over 23 days. The apparent increase of confluence between days 0 and 6 on the combination treatment arm is partially due to an increase in cell size rather than an increase in cell numbers (Supplementary Fig. S6).

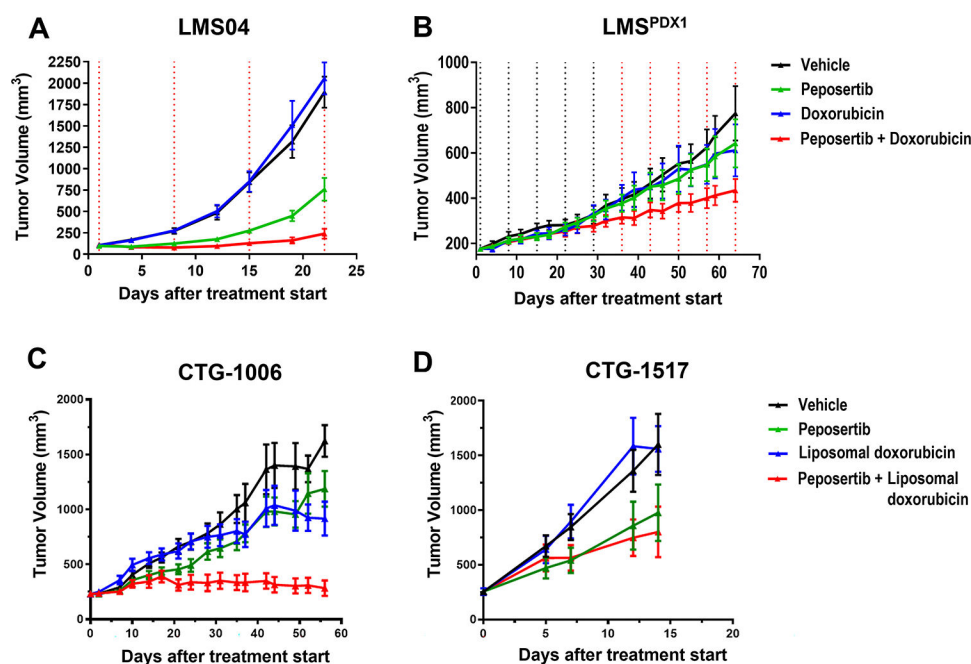


Figure 5:
In vivo effect of peposertib (100 mg/kg bid), low-dose doxorubicin (0.5–1 mg/kg qw, A and B) or low-dose pegylated liposomal doxorubicin (3 mg/kg qw, C and D) and combinations on LMS PDX with *TP53*, *RB1* and *ATRX* inactivation. **A.** LMS04 (dotted vertical red line denotes doxorubicin dose at 1 mg/kg). **B.** LMS^{PDX1} (dotted vertical line denotes doxorubicin dose, 0.5 mg/kg black line, and 1 mg/kg red line) **C.** CTG-1006. **D.** CTG-1517.